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Author One[a,c,1], Author Two[b,1,2], and Author Three[a]

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[1]A.O.(Author One) contributed equally to this work with A.T. (Author Two) (remove if not applicable).

[2]To whom correspondence should be addressed. E-mail: author.twoemail.com

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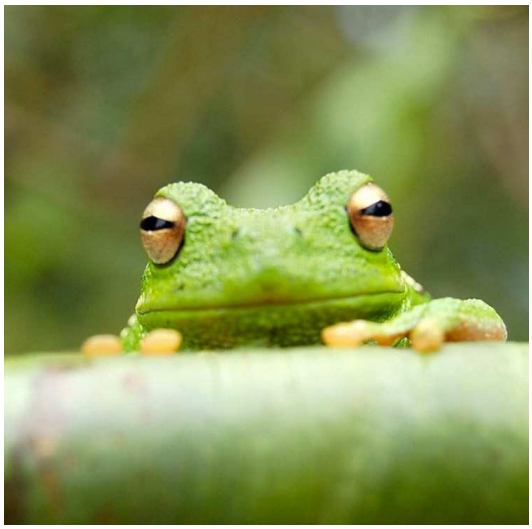

Fig. 1. Placeholder image of a frog with a long example legend to show justification setting.

Table 1. Comparison of the fitted potential energy surfaces and ab initio benchmark electronic energy calculations

Species	CBS	CV	G3
1. Acetaldehyde	0.0	0.0	0.0
2. Vinyl alcohol	9.1	9.6	13.5
3. Hydroxyethylidene	50.8	51.2	54.0

nomenclature for the TSs refers to the numbered species in the table.

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$$\begin{aligned}(x+y)^3 &= (x+y)(x+y)^2 \\ &= (x+y)(x^2+2xy+y^2) \\ &= x^3+3x^2y+3xy^3+x^3.\end{aligned} \quad [1]$$

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*Footnote Example 1

†Footnote Example 2

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Fig. 2. Placeholder image of a frog with a long example legend to show justification setting.

Table 2. Impact on Emission Behaviors by Socioeconomic Status

	(1)	(2)	(3)	(4)	(5)
	City-level		Firm-level	City-level	
Dep. Var.:	COD Emission (1,000 tons)		COD Emission (ton)	Firm Entry	Firm Exit
	Panel A: Population Share without College Education				
Share of Below College $\times$ Post_{05}	0.165***	0.292**	0.358***	0.623**	0.280
	(0.026)	(0.119)	(0.106)	(0.266)	(0.487)
Share of Below College	-0.208				
	(0.141)				
Post_{05}	-16.426***				
	(2.873)				
	Panel B: Population Share without High School Education				
Share of Below HS $\times$ Post_{05}	0.099***	0.218**	0.232**	0.453**	0.089
	(0.005)	(0.090)	(0.079)	(0.195)	(0.333)
Share of Below HS	-0.213*				
	(0.100)				
Post_{05}	-9.465***				
	(0.564)				

*** $\mathrm{P} < 0.01$, ** $\mathrm{P} < 0.05$, * $\mathrm{P} < 0.1$

Fig. 3. This legend would be placed at the side of the figure, rather than below it.

$$\begin{aligned}(x+y)^3 &= (x+y)(x+y)^2 \\ &= (x+y)(x^2+2xy+y^2) \\ &= x^3+3x^2y+3xy^3+x^3.\end{aligned} \quad [2]$$

Subsection for Method. Example text for subsection.

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1. M Belkin, P Niyogi, Using manifold stucture for partially labeled classification in *Advances in neural information processing systems.* pp. 929–936 (2002).
2. P Bérard, G Besson, S Gallot, Embedding riemannian manifolds by their heat kernel. *Geom. & Funct. Analysis GAFA* **4**, 373–398 (1994).
3. RR Coifman, et al., Geometric diffusions as a tool for harmonic analysis and structure definition of data: Diffusion maps. *Proc. Natl. Acad. Sci. United States Am.* **102**, 7426–7431 (2005).
4. RJF Thomas, "Enamel defects, well-being and mortality in a medieval Danish village," PhD thesis, Pennsylvania State University, University Park, PA (2003).
5. TM Toolan, "Advances in Sliding Window Subspace Tracking," M.Sc. thesis, University of Rhode Island, Kingston, RI (2005).

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